



BobSim

ISO4138 Vehicle Characterization

OpenModelica Executable with Modelica PI Control

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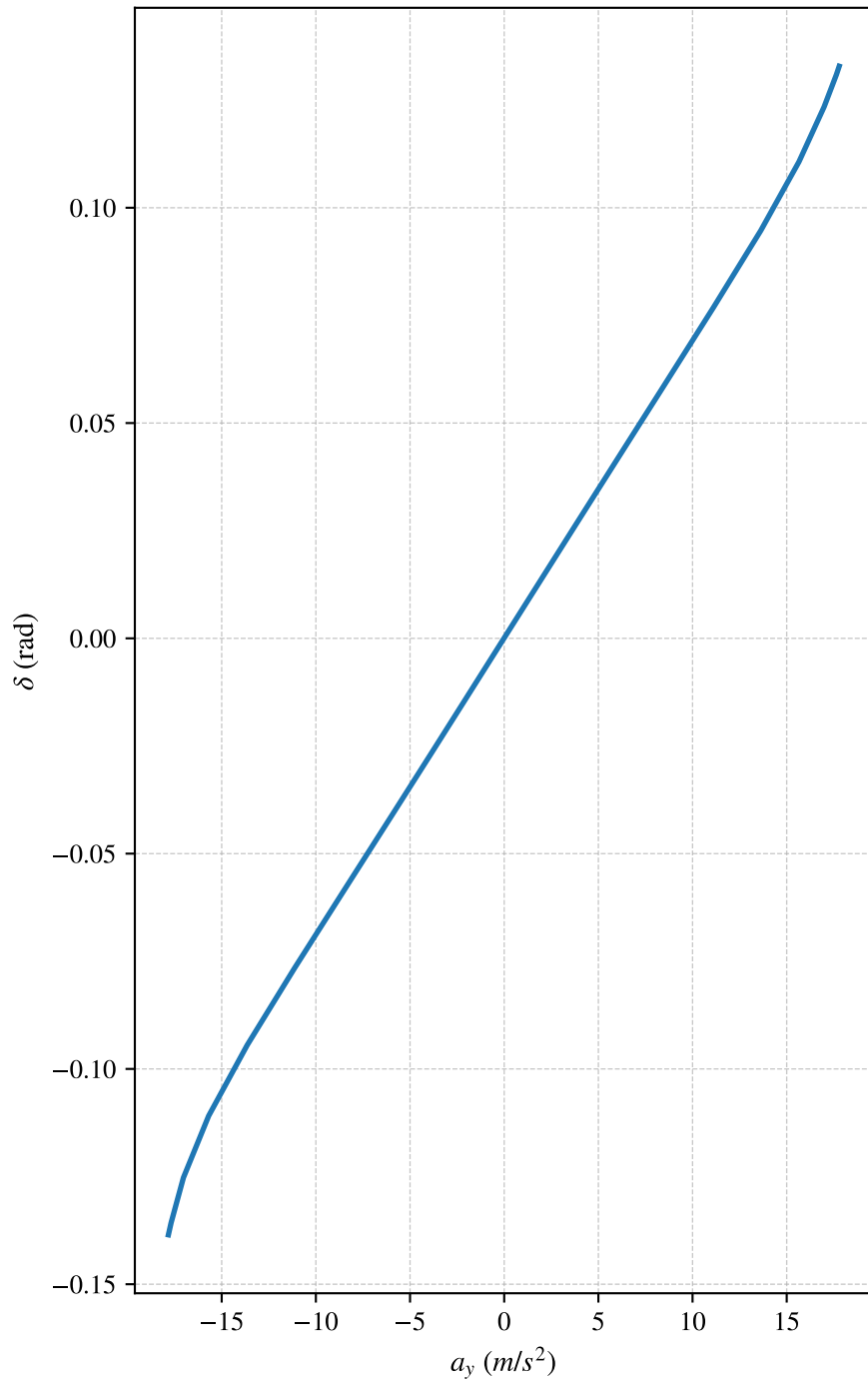
BobDyn

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- ISO4138 steady-state circular test performed
 - Constant velocity maintained at 15 m/s
 - Radius sweep from 12.5 m to 100 m
 - Tire model: MF52

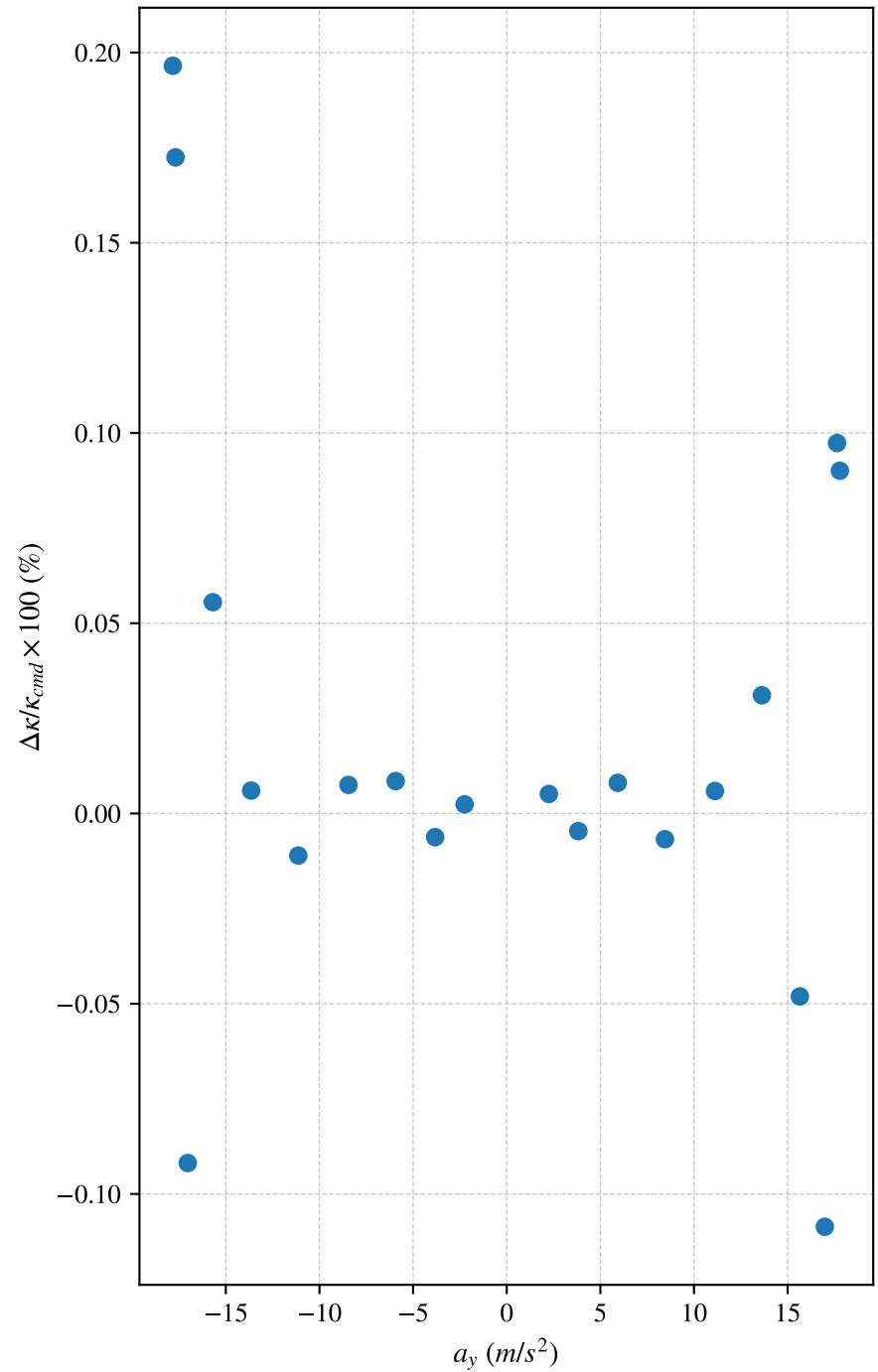
Metric	Value	Units
a_y Range	$-17.83 \rightarrow 17.79$	m/s^2
Understeer Gradient	0.00692	$\frac{\text{rad}}{\text{m/s}^2}$
	3.89	$\frac{\text{deg}}{g}$
Roll Gradient	1.228	$\frac{\text{deg}}{g}$
Max Radius Error	0.20	%

Steering Behavior vs a_y

Roadwheel Angle δ

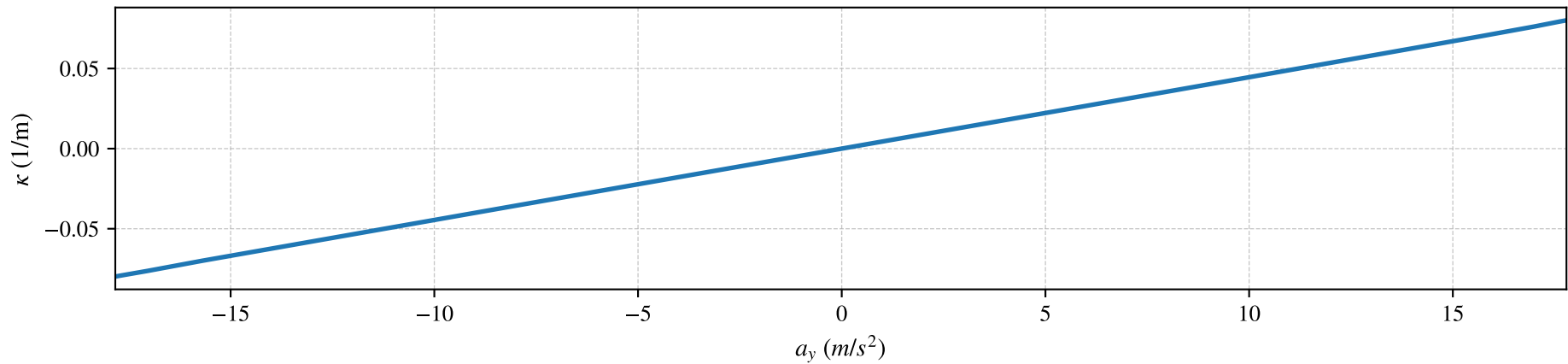


Curvature Tracking Error

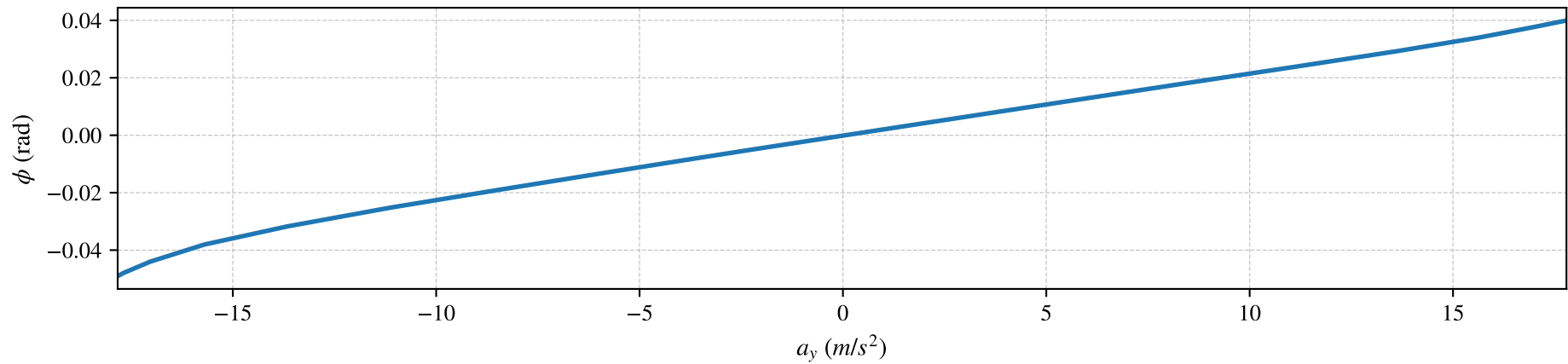


Vehicle States vs a_y

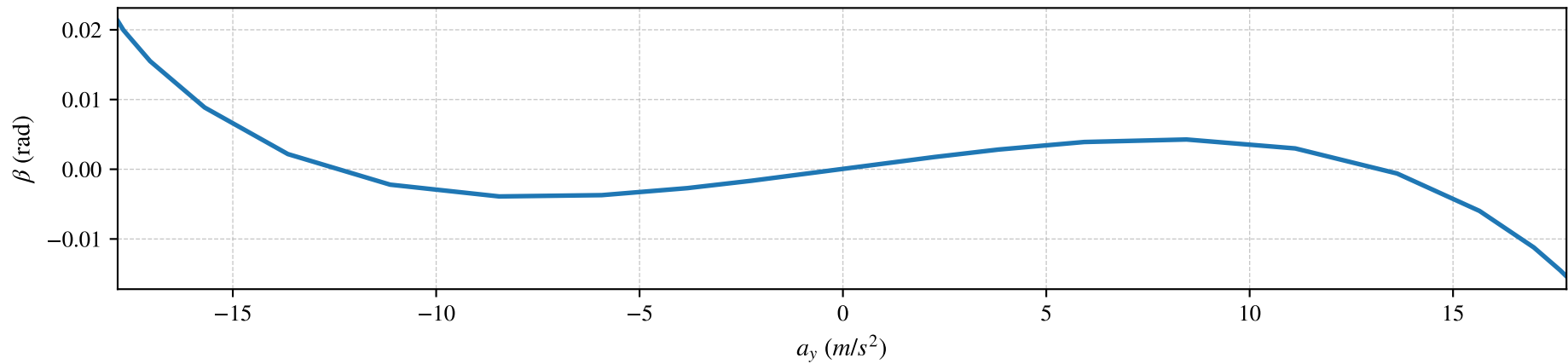
Curvature κ



Roll Angle ϕ

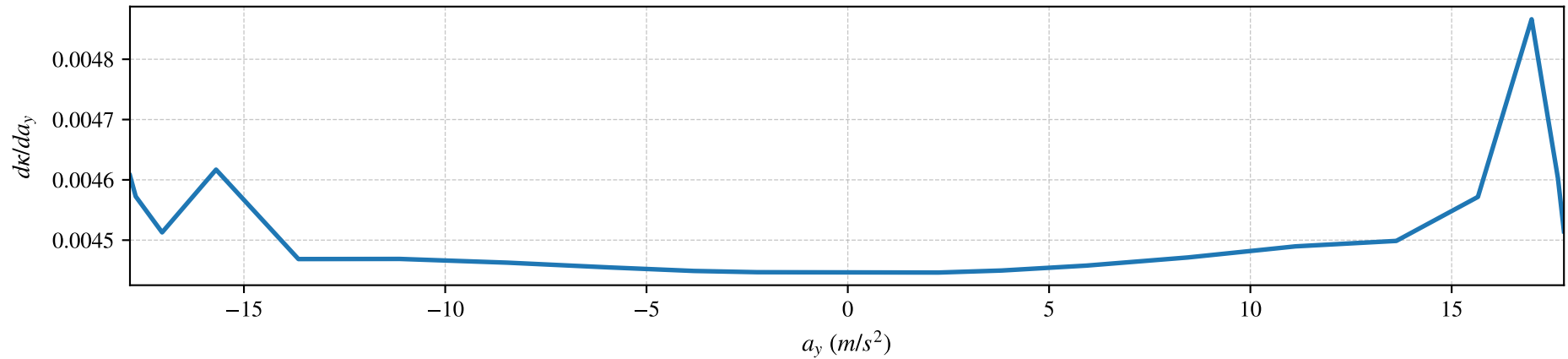


Sideslip β

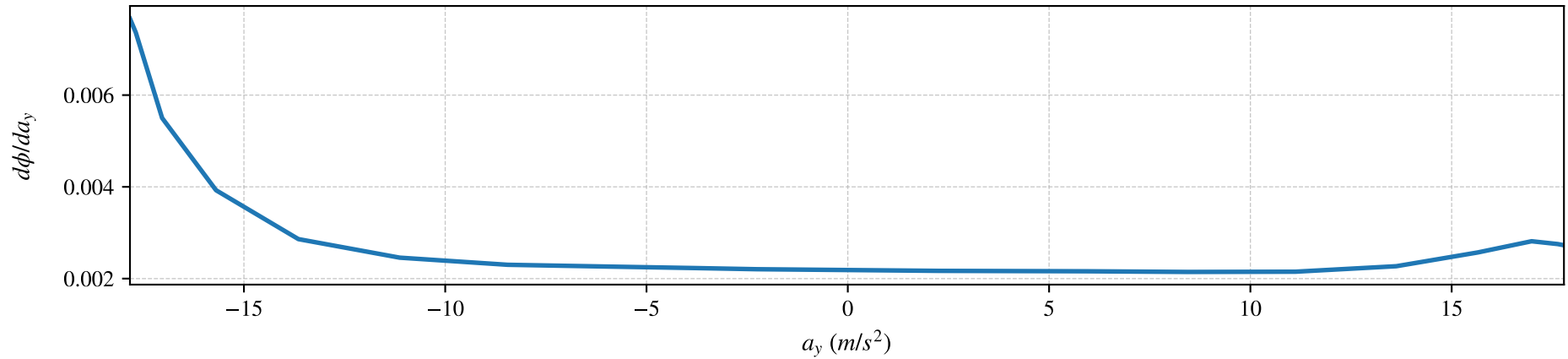


Response Sensitivities vs a_y

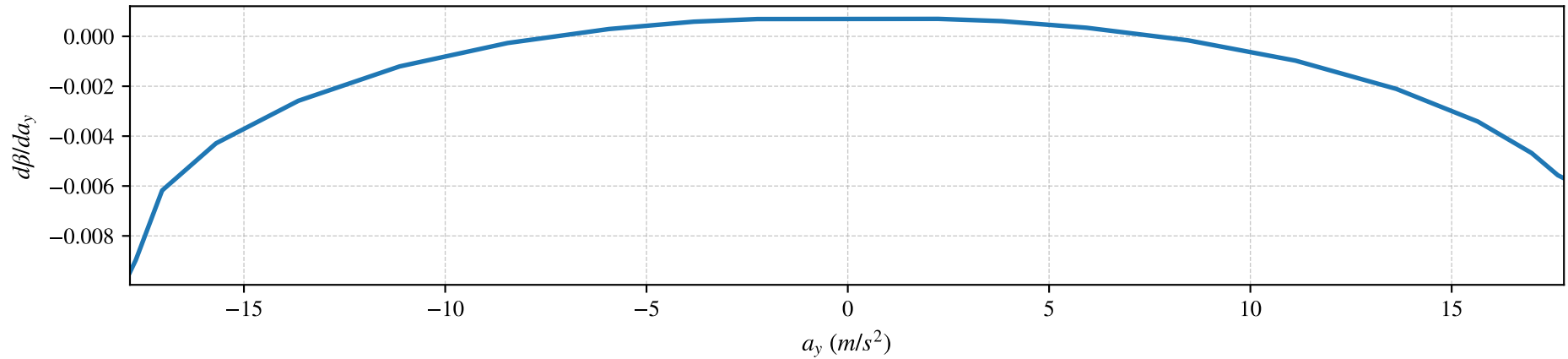
$$d\kappa/da_y$$



$$d\phi/da_y$$



$$d\beta/da_y$$



Steering Effort vs a_y

